

AHMAD JAMAL KHAN

LabVIEW Test-Automation Engineer | RF / Microwave Instrument Control & Calibration

Islamabad, Pakistan • Available remote — US CST hours • +92 301 8137118 • engr.ahmadjamalkhan@gmail.com
linkedin.com/in/ahmad-jamal-khan-jadoon • github.com/Skeletoskull

PROFESSIONAL SUMMARY

LabVIEW and VEE Pro test-automation engineer with 2.5+ years developing automated calibration and measurement systems for RF/microwave and sensor instrumentation. Specialize in writing clean, modular, well-documented instrument-control code and delivering validated test procedures on tight, ≤ 3 -day timelines. Currently build automated calibration programs for RF signal generators in Keysight VEE (VEE Pro) for a US test lab, driving bench instruments over SCPI/VISA and NI-DAQmx and validating every measurement live. Backed by a BS in Avionics Engineering with a strong microwave-measurement foundation. IELTS 7.5 (C1); proven in remote, cross-time-zone, live-validated workflows.

CORE SKILLS

Test Automation & Languages: LabVIEW (Advanced — OOP, Queued Message Handler / Actor Framework, modular plug-in architecture), Keysight/Agilent VEE (VEE Pro), Python (NumPy, SciPy, PyVISA), C/C++

Instrument Control: SCPI, VISA, NI-DAQmx, GPIB / USB / LAN; automated calibration, measurement validation, data logging & report generation

RF / Microwave Measurement: Signal Generators (automated calibration), Spectrum Analyzers, Power Sensors/Meters, S-parameters (S11 / return loss), RF link budgets; lab environment incl. VNAs, directional couplers and SOL standards

Instruments & Vendors: Keysight/Agilent, Tektronix, Rohde & Schwarz, Fluke, Siglent, Rigol; NI USRP (X310, B210), BladeRF, NI cRIO, DAQ

FPGA & Workflow: Verilog / VHDL, Intel Quartus, Xilinx Vivado; Git, AnyDesk / TeamViewer remote live-validation

PROFESSIONAL EXPERIENCE

Freelance LabVIEW / VEE Test-Automation Engineer — Remote (US Client)

Oct 2025 – Present

- Develop automated calibration programs for RF **signal generators in Keysight VEE (VEE Pro)** for a US microwave test lab, controlling bench instruments over SCPI/VISA.
- Deliver complete, validated test procedures in ≤ 3 days and prove measurement accuracy on live AnyDesk sessions, writing clean and maintainable code to lab standards.
- Logged 200+ billed hours on an ongoing long-term engagement, collaborating on a US CST schedule across instruments from Keysight, Tektronix, Rohde & Schwarz, Siglent and Rigol.

RF Systems Engineer — Shaheen Aero Traders (on-site, NASTP), Rawalpindi

Nov 2023 – Present

- Built a versatile **LabVIEW instrument-control library** to generate and measure complex RF waveforms (FM/AM/PSK/ASK/noise), verifying system performance with spectrum analyzers, signal generators and SDRs.
- Architected a distributed, multi-node SDR test system integrating 3x NI USRP X310, high-power amplifiers and antenna rotators across a 4-PC networked cluster.
- Developed a custom TCP/IP protocol for command-and-control plus UDP telemetry for real-time amplifier and rotator health monitoring.
- Designed a LabVIEW HMI for real-time spectrum visualization, measurement logging and automated system diagnostics.
- Built a multi-band GNSS resilience test suite (L1/L2/L5/S) to evaluate receiver performance under high-interference conditions; performed RF link-budget analysis and spectrum planning.
- Counter-UAS R&D: reverse-engineered DJI OcuSync 2.0 to decode drone telemetry using BladeRF and USRP.

Project Management Associate Intern — Excelerate — Remote (USA)

Mar – Apr 2025

- Co-developed a Quality-Assurance framework and supported onboarding and cross-team delivery for a global virtual hackathon.

Lecturer — NCS University System, Dept. of Management Sciences, Peshawar

Oct 2022 – Jan 2023

- Delivered lectures on C/C++ programming, operating systems and computer networking; designed and supervised practical labs.

Earlier: Engineering Intern — Pakistan Aeronautical Complex (PAC) Kamra, Avionics Production Factory (2021) • Intern — Cosmic Space Systems, Islamabad (2022).

SELECTED PROJECTS

Pneumatic Test Bench Suite V2 (LabVIEW 2021) — client project. Built a modular, plug-in-based test application (Queued Message Handler / actor architecture) acquiring from 8 pressure transducers and 8 flowmeters over NI-DAQmx (4–20 mA current-loop, continuous sampling), with a linear calibration module (gain/offset) and automated report generation with full error logging.

DroneSecurity — Windows Port (Open Source, Lead Developer) — ported a Linux DJI DroneID receiver to Windows 10/11 with BladeRF; implemented Zadoff-Chu sequence detection and Turbo decoding; authored 50,000+ words of documentation. github.com/Skeletoskull/DroneSecurity

DJI Drone Classifier — decoded DJI RemoteID on an NI USRP X310 with TwinRX daughterboards to expose live drone latitude/longitude, speed and serial number.

Smart Lead Enrichment System (Python) — asynchronous multi-source data-enrichment pipeline exposing REST API, web and CLI interfaces, with confidence scoring, cross-source verification and multi-tenant security.

BladeRF A4 FPGA Customization (Verilog) — modified the FPGA bitstream with custom GPIO/status logic to validate hardware-in-the-loop triggers.

EDUCATION

B.S. Avionics Engineering — Institute of Space Technology (IST), Islamabad 2018 – 2022
Relevant coursework: Microwave Engineering; Transmission Lines & Waveguides; Antenna Theory & Design; Analog & Digital Communication; Instrumentation & Measurements; Digital Signal Processing.

CERTIFICATIONS

- **Introduction to FPGA Design for Embedded Systems** — University of Colorado Boulder (Coursera), 2024
- **Hardware Description Languages for FPGA Design** — University of Colorado Boulder (Coursera), 2024
- Additional certifications in Project Management (Coursera) and Pakistan Engineering Council (PEC).

LANGUAGES

English — IELTS Academic 7.5 / CEFR C1 (Listening 7.5, Reading 8.5, Writing 6.5, Speaking 7.0) • **Urdu** — Native